

In this problem set, the questions marked with a (★) are required for students enrolled in 583 and bonus for students enrolled in 483.

1. A three-dimensional spacetime has the line element

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\phi^2. \quad (1)$$

- (a) (10 points) Find the explicit Lagrangian for the variational principle for geodesics in this spacetime in these coordinates.

Solution: Let $f(r) \equiv \left(1 - \frac{2M}{r}\right)$. Recall that the Lagrangian for geodesic motion is given by $\mathcal{L} = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}$ where λ is an affine parameter. Using the metric components, we obtain

$$\mathcal{L} = \left[f(r) \dot{t}^2 - f(r)^{-1} \dot{r}^2 - r^2 \dot{\phi}^2 \right]^{1/2}. \quad (2)$$

where the overdot is a derivative with respect to λ .

- (b) (10 points) Using the results of (a) write out the components of the geodesic equation by computing them from the Lagrangian.

Solution: The Euler–Lagrange equations,

$$\frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0,$$

yield the geodesic equations.

(i) **t –equation:** Since $\mathcal{L}$ is independent of t ,

$$\frac{d}{d\lambda} \left(\frac{1}{\mathcal{L}} f \dot{t} \right) = 0,$$

which gives

$$\ddot{t} + \frac{f'(r)}{f(r)} \dot{r} \dot{t} = 0. \quad (3)$$

where now the overdot is a derivative with respect to τ and we have used the fact that $\mathcal{L} = d\tau/d\lambda$.

(ii) **ϕ –equation:** Similarly, since $\mathcal{L}$ is independent of ϕ ,

$$\frac{d}{d\lambda} \left(\frac{1}{\mathcal{L}} r^2 \dot{\phi} \right) = 0,$$

or

$$\ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} = 0. \quad (4)$$

(iii) **r -equation:** For the radial coordinate we find

$$\ddot{r} + \frac{1}{2}f f' \dot{t}^2 - \frac{1}{2}\frac{f'}{f}\dot{r}^2 - fr\dot{\phi}^2 = 0, \quad (5)$$

where

$$f'(r) = \frac{2M}{r^2}.$$

- (c) (5 points) Read off the nonzero Christoffel symbols for this metric from your results in (b).

Solution: Comparing the above equations with the geodesic equation

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0,$$

we can read off the nonzero Christoffel symbols:

$$\Gamma_{tr}^t = \Gamma_{rt}^t = \frac{f'}{2f}, \quad (6)$$

$$\Gamma_{tt}^r = \frac{1}{2}ff', \quad (7)$$

$$\Gamma_{rr}^r = -\frac{1}{2}\frac{f'}{f}, \quad (8)$$

$$\Gamma_{\phi\phi}^r = -fr, \quad (9)$$

$$\Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = \frac{1}{r}. \quad (10)$$

2. **Christoffel Symbols, aka connection coefficients** Consider a change of basis $\tilde{e}_\mu = A_\mu^\nu e_\nu$.

- (a) (10 points) Show that the components of the Christoffel symbol in the new basis are related to the components in the old basis by

$$\tilde{\Gamma}_{\rho\nu}^\mu = A_\rho^\sigma (A^{-1})_\tau^\mu [A_\nu^\lambda \Gamma_{\sigma\lambda}^\tau + \partial_\sigma A_\nu^\tau]. \quad (11)$$

Given the above equation, are the Christoffel symbols tensors? Why or why not?

Solution: Let's start with

$$\begin{aligned}
\tilde{\Gamma}_{\rho\nu}^{\mu} \tilde{e}_{\mu} &= \nabla_{(A_{\rho}^{\sigma} e_{\sigma})} A_{\nu}^{\lambda} e_{\lambda} \\
&= A_{\rho}^{\sigma} \nabla_{\sigma} (A_{\nu}^{\lambda} e_{\lambda}) \\
&= A_{\rho}^{\sigma} (\partial_{\sigma} A_{\nu}^{\lambda} e_{\lambda} + \Gamma_{\sigma\lambda}^{\tau} A_{\nu}^{\lambda} e_{\tau}) \\
&= A_{\rho}^{\sigma} e_{\lambda} \partial_{\sigma} A_{\nu}^{\lambda} + A_{\rho}^{\sigma} \Gamma_{\sigma\lambda}^{\tau} A_{\nu}^{\lambda} e_{\tau} \\
&= (A_{\rho}^{\sigma} \partial_{\sigma} A_{\nu}^{\tau} + A_{\rho}^{\sigma} \Gamma_{\sigma\lambda}^{\tau} A_{\nu}^{\lambda}) e_{\tau} \\
&= (A_{\rho}^{\sigma} \partial_{\sigma} A_{\nu}^{\tau} + A_{\rho}^{\sigma} \Gamma_{\sigma\lambda}^{\tau} A_{\nu}^{\lambda}) (A^{-1})_{\tau}^{\mu} \tilde{e}_{\mu}.
\end{aligned} \tag{12}$$

We can cancel out the common $\tilde{e}_{\mu}$ on both sides to find that

$$\begin{aligned}
\tilde{\Gamma}_{\rho\nu}^{\mu} &= (A_{\rho}^{\sigma} \partial_{\sigma} A_{\nu}^{\tau} + A_{\rho}^{\sigma} \Gamma_{\sigma\lambda}^{\tau} A_{\nu}^{\lambda}) (A^{-1})_{\tau}^{\mu} \\
&= A_{\rho}^{\sigma} (A^{-1})_{\tau}^{\mu} [A_{\nu}^{\lambda} \Gamma_{\sigma\lambda}^{\tau} + \partial_{\sigma} A_{\nu}^{\tau}].
\end{aligned} \tag{13}$$

Note that the first term corresponds to the transformation law of tensors but the second term depends on derivatives of the Jacobian A . Therefore, the Christoffel symbols do not obey the tensor transformation law and are not tensors.

- (b) (10 points) Does the difference of two connections, $S_{\rho\nu}^{\mu} = (\Gamma_1)_{\rho\nu}^{\mu} - (\Gamma_2)_{\rho\nu}^{\mu}$ obey the tensor transformation law?

Solution: From the previous part, we learned that the connections transform as

$$\tilde{\Gamma}_{\rho\nu}^{\mu} = A_{\rho}^{\sigma} (A^{-1})_{\tau}^{\mu} [A_{\nu}^{\lambda} \Gamma_{\sigma\lambda}^{\tau} + \partial_{\sigma} A_{\nu}^{\tau}], \tag{14}$$

where the first term corresponds to the transformation law of tensors but the second term does not. However, if we take the transformation of the difference of two different connections,

$$\begin{aligned}
S_{\rho\nu}^{\mu} &= A_{\rho}^{\sigma} (A^{-1})_{\tau}^{\mu} [A_{\nu}^{\lambda} (\Gamma_1)_{\sigma\lambda}^{\tau} + \partial_{\sigma} A_{\nu}^{\tau}] \\
&\quad - A_{\rho}^{\sigma} (A^{-1})_{\tau}^{\mu} [A_{\nu}^{\lambda} (\Gamma_2)_{\sigma\lambda}^{\tau} + \partial_{\sigma} A_{\nu}^{\tau}] \\
&= A_{\rho}^{\sigma} (A^{-1})_{\tau}^{\mu} A_{\nu}^{\lambda} [(\Gamma_1)_{\sigma\lambda}^{\tau} - (\Gamma_2)_{\sigma\lambda}^{\tau}],
\end{aligned} \tag{15}$$

the second term cancels and we are left with the transformation law for tensors.

3. (15 points) (★) An observer in a circular orbit of circumference $2\pi R$ around a charged, spherical, black hole (Reissner-Nordström black hole) of mass M and charge q measures local electric and magnetic fields. What are their strengths and orientations? The Reissner-Nordström metric is given by

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \tag{16}$$

where $f(r) \equiv 1 - 2GM/r + q^2/r$.

Solution: First, we need to find the angular velocity $\Omega = d\phi/dt$ of circular orbits in such a geometry. We start with the geodesic equations, $d^2x^\mu/d\tau^2 + \Gamma_{\alpha\beta}^\mu u^\alpha u^\beta$, for which we need to solve for the Christoffel symbols. The ones of interest (setting $\theta = \pi/2$) are

$$\begin{aligned}\Gamma_{rtt} &= -g_{tt,r} = \frac{GM}{r^2} - \frac{q^2}{r^2} \\ 2\Gamma_{rt\phi} &= -g_{t\phi,r} = 0 \\ \Gamma_{r\phi\phi} &= -\frac{1}{2}g_{\phi\phi,r} = -r.\end{aligned}\tag{17}$$

Thus, we find that for the orbiting observer

$$\Omega^2 R - \left(\frac{GM}{R^2} - \frac{q^2}{R^3} \right) = 0 \implies \Omega = \pm \left(\frac{GM}{R^3} - \frac{q^2}{R^4} \right)^{1/2}.\tag{18}$$

The proper velocity of the orbiting observer relative to a coordinate stationary observer is

$$\hat{v} = \frac{d\hat{\phi}}{d\hat{t}} = \frac{R}{f(R)^{1/2}} \Omega = \left(\frac{MR - q^2}{R^2 - 2MRq^2} \right)^{1/2},\tag{19}$$

where we indicate the orthonormal components in the stationary frame with $\hat{\cdot}$. In the stationary frame, we can calculate the local electromagnetic field from undergraduate E&M:

$$\hat{E}_{\hat{r}} = \frac{q}{R^2}.\tag{20}$$

Note that since the observer is stationary (as is the BH), there is no magnetic field. To get the fields in the frame of the orbiting observer, we Lorentz transform the above and apply the standard relations for E&M fields to find that

$$\begin{aligned}E_{\hat{r}} &= (q - \hat{v}^2)^{-1/2} \frac{q}{R^2} = \frac{q}{R^2} \left(\frac{R^2 - 2MR + q^2}{R^2 - 3MR + 2q^2} \right)^{1/2} \\ B_{\hat{\theta}} &= (q - \hat{v}^2)^{-1/2} \hat{v} \frac{q}{R^2} = \frac{q}{R^2} \left(\frac{MR - q^2}{R^2 - 3MR + 2q^2} \right)^{1/2}.\end{aligned}\tag{21}$$